BCS-077	Software Quality Management	L T P 3 0 0	3 Credits

Pre-requisites: None
Course Objectives:

CO1	To make the students understand the quality management process in the development of software.
CO2	To make the students understand the importance of standards in the quality assurance process and their impact on the final product.
CO3	To understand quality control and reliability
CO4	To study ISO 9000 standard
CO5	To learn about CMM and CMMI – Six Sigma concepts.

Detailed Syllabus

MODULE - I

INTRODUCTION TO SOFTWARE QUALITY:
Software Quality – Hierarchical models of Boehm and McCall – Quality measurement – Metrics measurement and analysis – Gilb's approach – GQM Model
SOFTWAREQUALITY ASSURANCE: Quality tasks – SQA plan – Teams – Characteristics – Implementation – Documentation – Reviews and Audits

MODULE - II

QUALITY CONTROL AND RELIABILITY:
Tools for Quality – Ishikawa's basic tools – CASE tools – Defect prevention and removal – Reliability models – Rayleigh model – Reliability growth models for quality assessment

MODULE - III

QUALITY MANAGEMENT SYSTEM:
Elements of QMS – Rayleigh model framework – Reliability Growth models for QMS – Complexity metrics and models – Customer satisfaction analysis.
Quality Standards: Need for standards – ISO 9000 Series – ISO 9000-3 for software development – CMM and CMMI – Six Sigma concepts.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Distinguish between the various activities of quality assurance, quality planning and quality control
CO2	Aware of Ishikawa's basic tools
CO3	Familiar with quality standards
CO4	Familiar with basic tools and models for quality assessment
CO5	Understand about CMM and CMMI – Six Sigma concepts.

Text Books:

1. Allan C. Gillies, "Software Quality: Theory and Management", Thomson Learning, 2003.
2. Stephen H. Kan, "Metrics and Models in Software Quality Engineering", Pearson Education (Singapore) Pte Ltd., 2002.

References Books:

1. Norman E. Fenton and Shari Lawrence Pfleeger, "Software Metrics" Thomson, 2003
2. Mordechai Ben – Menachem and Garry S.Marliss, "Software Quality", Thomson Asia Pte Ltd, 2003.
3. Mary Beth Chrissis, Mike Konrad and Sandy Shrum, "CMMI", Pearson Education (Singapore) Pte Ltd, 2003.
4. ISO 9000-3 "Notes for the application of the ISO 9001 Standard to software development".